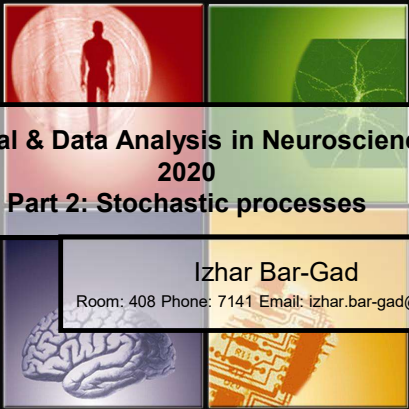



**Signal & Data Analysis in Neuroscience
2020**

Part 2: Stochastic processes

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Overview

- Stochastic processes
- Point processes
- Appendix: extracellular recording

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Modeling our measurements

- Repetition of the same experiment will not lead to the exact same response.
- **Example:** The spike train of a neuron in response to stimuli is different...
- Typically we would like to know:
 - When is something unexpected?
 - What are the "normal" values?
- Thus, analyzing a sequence of measurements requires modeling of the underlying process.

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Stochastic process

Definition

- **Stochastic** – (1) Involving chance or probability (2) Random (3) Non-deterministic (Merriam-Webster & Wikipedia).
- **Stochastic process** - an indexed collection of random variables $\{X_i\}$, where the index i ranges through an index set I , defined on the probability space (Ω, P) . The index set may be discrete or continuous (Wikipedia).

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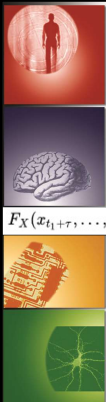


Stochastic process

Special cases

- A stochastic process defined over the time interval domain is called a **time series**.
 - Example of a continuous time series: temperature in BIU throughout the day.
 - Example of a discrete time series: amount of rain in BIU on a specific day on of the year.
 - Example of a quantized discrete time series: did rain fall in BIU on a specific day of the year?
 - Example of discrete non-time series: height of people entering Gonda building each day
- A stochastic process defined over the space interval domain is called a **random field**.

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Stationary processes

Strict / Strong

- A stochastic process whose unconditional joint probability distribution does not change when shifted in its index (typically time).
$$F_X(x_{t_1+\tau}, \dots, x_{t_n+\tau}) = F_X(x_{t_1}, \dots, x_{t_n}) \quad \text{for all } \tau, t_1, \dots, t_n \in \mathbb{R} \text{ and for all } n \in \mathbb{N}$$
- Probability density function (PDF) – $p(x)$ describes the distribution of a continuous random variable, x .
$$\int_{-\infty}^{\infty} p(x) dx = 1$$
- Cumulative function: $F(x) = \int_{-\infty}^x p(y) dy$
- Survival function : $1 - F(x) = \int_x^{\infty} p(y) dy$
- N^{th} order stationary process is defined for all $n=\{1, \dots, N\}$

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Stationary processes

Wide-sense / Weak

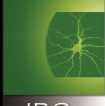
- A weak or wide-sense stationary (WSS) process only is a second degree stationary process.
- Equal mean value

$$E(X(t)) = \mu_x(t) = \mu_x(t + \tau) \quad \forall \tau \in \mathbb{R}$$
- Covairance dependent only on index difference

$$E(X(t_1) - \mu_x(t_1) \cdot (X(t_2) - \mu_x(t_2))) = Cov(X(t_1), X(t_2))$$

$$= Cov(X(t_1 + \tau), X(t_2 + \tau)) = Cov(X(t_1 - t_2), 0)$$






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Stationary processes - examples

- Stationary example:** Sequence of L/R button presses. Each press has a 90% probability of being in the same direction as its predecessor.
 - Stationary despite strong temporal covariance.
- Non-stationary example:** Amount of rainfall for each day of the year.
 - In many cases long term changes may be removed using **de-trending** techniques.






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Ergodicity

- If averaging over time and space are equal the process is **ergodic**.
- Ergodicity is usually described in terms of properties of an ensemble of objects.
- Example:** Finding out how people spend their spare time. Sampling one person over 1000 days would yield the same result as sampling 1000 people once in an ergodic system.

Reading material:
<http://news.softpedia.com/news/What-is-ergodicity-15686.shtml>






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Ergodic & stationary processes

- In an ergodic process, the following are equal
 - Averaging across repeated trials
 - Averaging across time for a single trial
- An ergodic process is always stationary, the reverse may not be true
- A stationary process is ergodic if samples that are far enough in time are independent (asymptotic independence).

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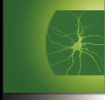

Overview

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- Appendix: extracellular recording

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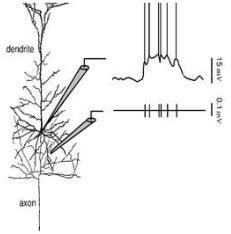
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Intracellular vs. Extracellular neuronal potentials

- Intracellular soma
- Extracellular



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Extracellular spike trains

- Transformation from a continuous recording to a series of discrete **timestamps**.
- Is all the information contained in the **timing** of the spikes?
- What are we losing?
 - Spike shapes
 - Non spiking activity
 - Sub-threshold activity

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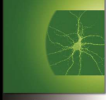

Time series & Point processes

- Continuous time series
 - Electroencephalogram (EEG)
 - Electromyogram (EMG)
 - Intracellular potential
 - ...
- (Note: "Continuous" is the common term but is misleading since it applies to both discrete and continuous in time)
- Stochastic point processes
 - Neural action potentials
 - Heart beats
 - Behavioral events
 - ...

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Delta functions (reminder)

- Dirac's delta function

$$\delta(x - \tau) = 0 \quad x \neq \tau$$

$$\int_{-\infty}^{\infty} \delta(x - \tau) dx = 1$$
- Kronecker's delta function

$$\delta(n - k) = \begin{cases} 1 & n = k \\ 0 & n \neq k \end{cases} \quad \sum_{n=-\infty}^{\infty} \delta(n) = 1$$

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Point process

- The spike train is represented by the sum of Dirac's delta functions at its firing times (t_i)

$$\rho(t) = \sum_{i=1}^n \delta(t - t_i)$$

- Point processes are unitary events in time. The actual values in time are meaningless.

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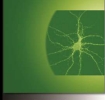

Properties of a single spike train

- Firing rate
- Response to events
- Firing pattern
- Exact timing
- Entropy
- ...

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The neural transformation

$x(t) \rightarrow \text{Sensors} \rightarrow r(t) \rightarrow \text{Spike Generator} \rightarrow \rho(t)$

$x(t)$ = external signal
 $r(t)$ = spike rate
 $\rho(t)$ = actual spikes

We observe $\rho(t)$, and we need to estimate $r(t)$
 (eventually we will use this to estimate $x(t)$)

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Firing rate definition

- There are different definitions to firing rate
 - r – rate over the whole period T also called *spike count rate*
 - $\langle r \rangle$ - rate averaged over all the trials, also called *average firing rate*
 - $r(t)$ – trial average rate over a short period ($\Delta t \rightarrow 0$)

and they are constantly mixed...

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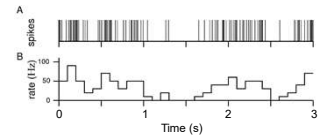





Firing rates – number of spikes

Firing rate: $r = \frac{n}{T} = \frac{1}{T} \int_0^T \rho(\tau) d\tau$.

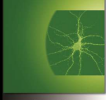
$\Delta t = 100$ ms



From: Theoretical neuroscience / Dayan & Abbott

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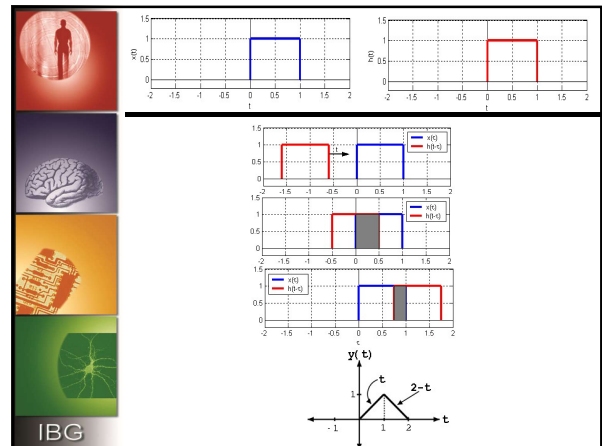


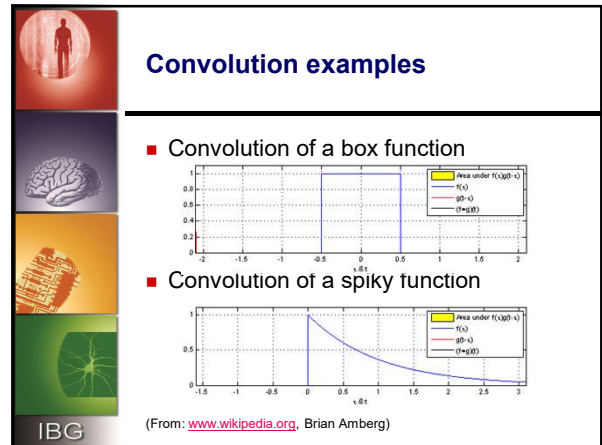
Convolution

- Convolution is an **operator** which takes two functions f and g and produces a third function that represents the overlap between f and a reversed version of g .
- Continuous: $(f * g)(t) = \int f(\tau)g(t - \tau) d\tau$
- Discrete: $(f * g)(m) = \sum_n f(n)g(m - n)$

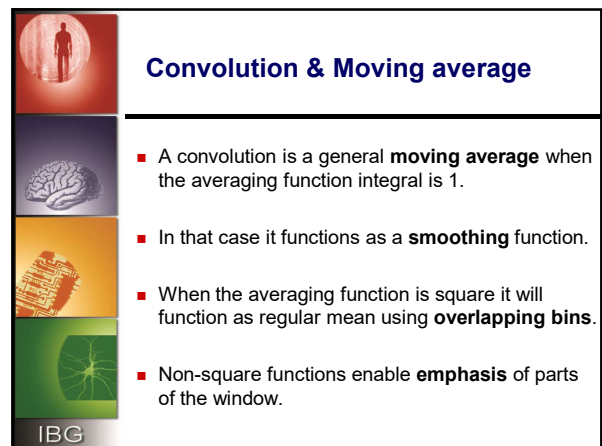
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Firing rates – sliding windows

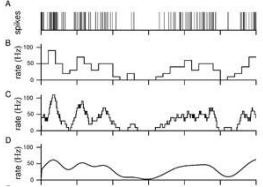
Other definitions of firing rate use a sliding window: $r(t) = \int_{-\infty}^{\infty} dw(\tau)p(t-\tau)$, with

$$w(t) = \frac{1}{\Delta t}, \quad -\Delta t/2 \leq t \leq \Delta t/2$$

$$w(t) = \frac{1}{\sqrt{2\pi}\sigma_w} \exp\left(-\frac{t^2}{2\sigma_w^2}\right)$$

Sliding rectangular window
 $\Delta t = 100$ ms

Sliding Gaussian window
 $\sigma_1 = 100$ ms

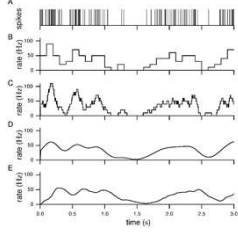


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Firing rates - causal windows

Temporal averaging with windows is non-causal. A causal alternative is $w(t)=[\alpha^2 t e^{-\alpha t}]_+$.



Sliding causal window
 $1/\alpha = 100$ ms

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Smoothing & Convolution

- Smoothing and convolution pitfalls
 - Introduces spurious correlations over time
 - Hidden assumption about smoothness of the external sensory or motor data
 - Edge effects: what happens at the start and end of the data?
 - Phase lag: peaks of smoothed data may occur later than the peaks in the original data. True for non-symmetric kernels and all causal filters

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Tuning curves

- $r()$ can be a function of something other than time. e.g. $r(\text{angle})$ if the rate varies with direction of hand movement
- $r(x)$ will still be time-varying if the argument x changes with time as $x(t)$. It can also change dependence on x , if $r = r(x, t)$.
- Describes the "tuning" of the neuron. x can be a scalar, vector, or function (pattern)
- A tuning curve is a model for the neuron's behavior, and is always an approximation since neurons are likely to have multiple inputs and respond to multiple internal and external variables.



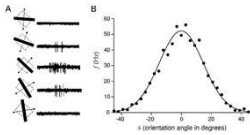
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Sensory tuning curves

- For sensory neurons, the firing rate depends on the stimulus s
- Extra cellular recording V1 monkey
- Response depends on angle of moving light bar
- Average over trials is fitted with a Gaussian

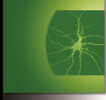


$$r(s) = r_{\max} \frac{(s - s_{\max})^2}{2 \cdot \sigma^2_{\max}}$$

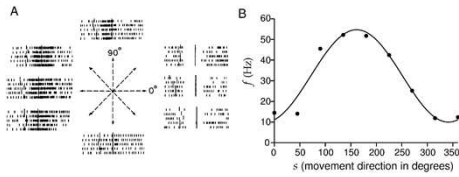


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Motor tuning curves



Extra cellular recording of monkey primary motor cortex M1 in arm-reaching task. Average firing rate is fitted with $r(s) = r_0 + (r_0 \cos(\max))$



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Spike count variability

- Tuning curves model average behavior.
- Deviations of individual trials are given by a noise model.
 - Additive noise is independent of stimulus
 $r(s) = f(s) + \xi$
 - Multiplicative noise is proportional to stimulus
 $r(s) = f(s) + g(s) \cdot \xi$

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Signal to noise ration

- **Power** of the signal (mean square)

Discrete $\frac{1}{N} \sum_{i=1}^N x_i^2$ Continuous $\frac{1}{T} \int_0^T x_i^2$
- **Amplitude** of the signal (root mean square)

Discrete $\sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$ Continuous $\sqrt{\frac{1}{T} \int_0^T x_i^2}$
- The signal to noise ratio (SNR) may be calculated directly: $\frac{ms(signal)}{ms(noise)}$ or $\left(\frac{rms(signal)}{rms(noise)}\right)^2$
- However, typically a decibel (dB) scale is used...

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Decibel (dB)

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- The **decibel (dB)** is a logarithmic measure of the ratio between two quantities:

$$SNR_{dB} = 10 \log_{10} \frac{ms(signal)}{ms(noise)} \text{ or } 20 \log_{10} \frac{rms(signal)}{rms(noise)}$$
- SNR of 3 dB is roughly double the power while 10 dB is ten times the power.
- SNR of 6 dB is roughly double the amplitude while 20 dB is ten times the power.

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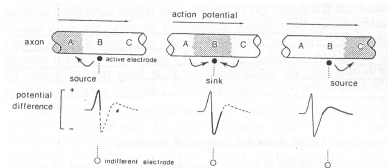
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Formation of extracellular potential I

- Different membrane potentials of the neuron lead to flow of current within the neuron which is matched by an **extracellular return current**.

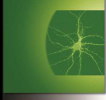


Sink – Active area, current flows into the neuron
Source – Inactive region, current flow out of the neuron.

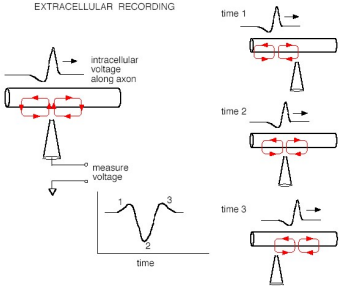


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Formation of extracellular potential II



Axon – Triphasic shape (+/-/+)



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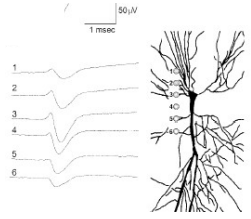





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Somatic extracellular spike shape

- Soma – Biphasic shape (-/+),
 - Negative due to flow from the initial segment
 - Positive due to flow to dendrite.
- Magnitude is proportional to surface area divided by the distance.



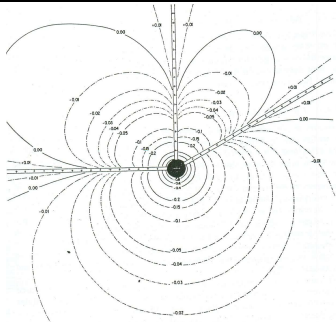
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Decay of extracellular signal



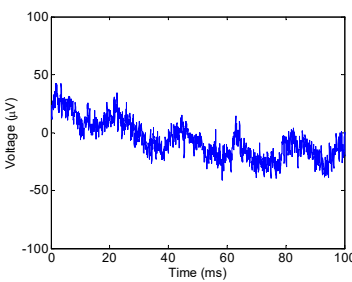
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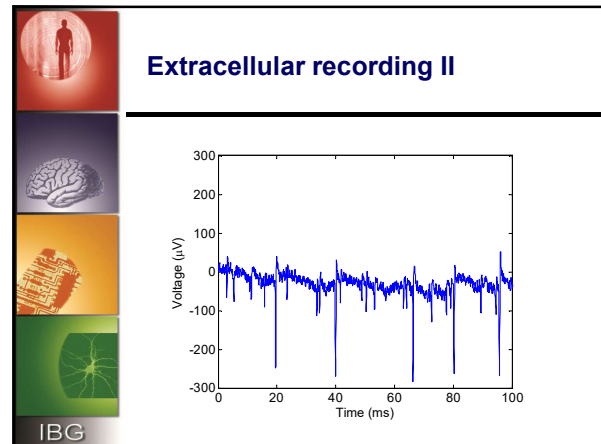



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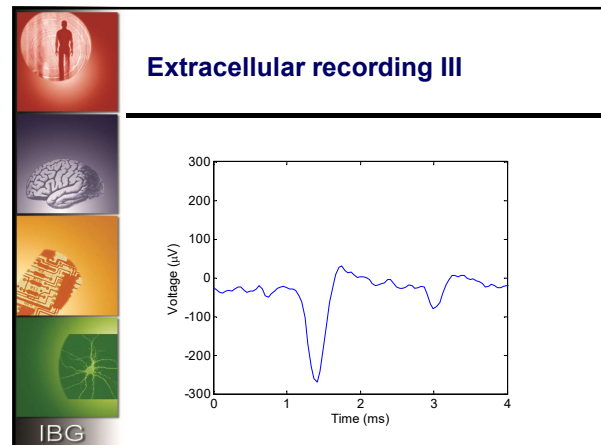
Extracellular recording I



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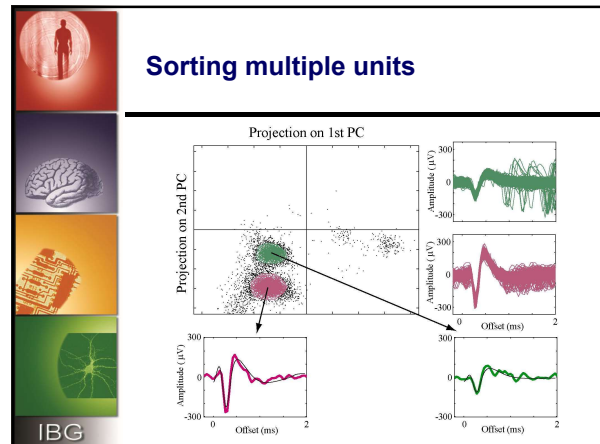
Multiple units

- The electrode detects multiple neurons (also called **units**) which are close to its tip.
- The signals differ in:
 - **Amplitude** - dependent on cell size and distance.
 - **Phase shape** - depends on direction to soma, axon & dendrites.
 - **Temporal shape** - dependent on cell type.
- Spikes from the same neuron also vary significantly due to noise, bursts, drift of electrode, etc..

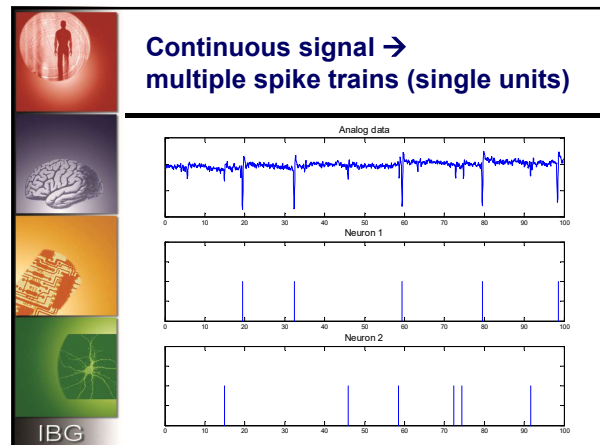
The diagram on the right shows a cross-section of neurons with an electrode tip. Arrows indicate the direction of signal propagation from different parts of the neurons (soma, axon, dendrites) towards the electrode, illustrating how the recorded signal's amplitude and phase shape vary based on the electrode's proximity to different cellular components.

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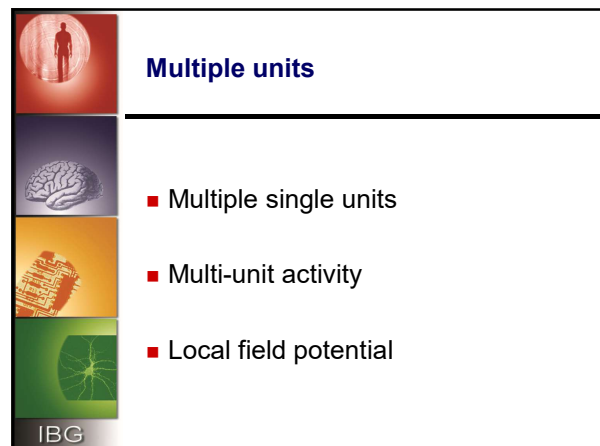
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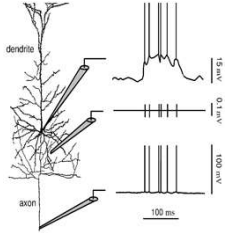
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Intracellular vs. Extracellular neuronal potentials

- Intracellular soma
- Extracellular
- Intracellular axon



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Spike trains

- Transformation from a continuous recording to a series of discrete **timestamps**.
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